

Pooling Contracts between Venture Capital Institutions

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Outline

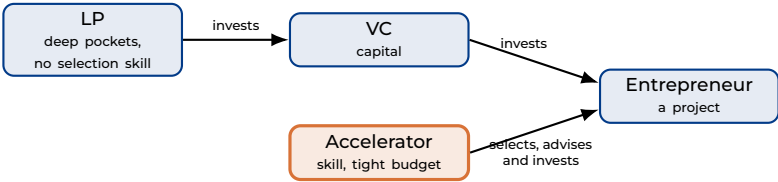
- 1 Introduction
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- 3 Contribution
- 4 Model
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Who does what in venture capital

The classic chain



With an accelerator



Contracts between VC institutions

- **VCs ↔ Limited Partners.**

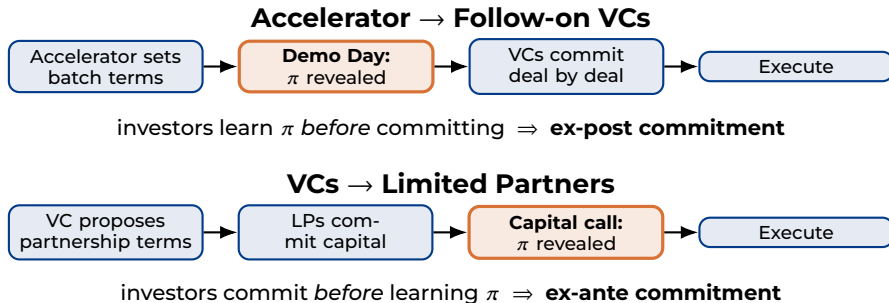
- **Ex-ante commitment.** LPs commit before seeing any target company, in 93% of deals (*Lerner et al., 2022*).
- **Pooling investment split.** Fixed at 1–5% of the total cost of investment.
- **Pooling return split.** Fixed at 20% of the return plus 2% of committed capital.

- **Accelerators ↔ Follow-on VCs.**

- **Ex-post commitment.** A demo day reveals the batch to follow-on VCs before they engage.
- **Pooling.** Terms are posted in advance and identical within a batch:

	Fixed amount, fixed share	Contingent tranche
Y Combinator	\$125K for 7%	\$375K uncapped MFN safe, 2022–
Techstars	\$20K for 6%, 5% since 2025	\$100K note capped at \$3M, then MFN
500 Global	\$150K for 6%	—
Entrepreneur First	\$125K for 8%, London	\$125K uncapped MFN safe
Antler	£125K for 8.5%, UK	£85K convertible note
	↑ this paper	↑ job market paper

Two relationships, two timings



- **Performance of contracts between VCs and LPs.**

- Lerner and Nanda (2020); Metrick and Yasuda (2010): Management fees incentivize funds to grow at the expense of performance.
- Axelson et al. (2009); Fang et al. (2015): Blind pool contracts minimizes VCs' incentives to overstate projects' values.
- Gourié et al. (2024); Lerner and Nanda (2020): LP's demand for VC is high (and concentrated).

- **Emergence and performance of the accelerator model.**

- Ewens, Nanda and Rhodes-Kropf (2018): Cheaper experiments move early-stage investors to smaller initial stakes in more startups.
- Gonzalez-Urbe and Leatherbee (2017): Increases VC fundraising and scale.

- **Theory of contracts.**

- Myerson (1983): A common mechanism can be proposed by an informed principal whatever the specific projects encountered.
- Maskin and Tirole (1992); Myerson (1983); Holmstrom and Myerson (1983); Guesnerie and Laffont, (1984); Kamien and Schwartz, (1991)

This paper

A first analysis of contracts between venture capital institutions, in an informed-principal framework with competitive agents.

Striking findings

- Under both regimes, pooling contracts are strategically optimal for the agency.
- Under ex-post commitment:
 - the unique optimal contract is pooling,
 - the agency's budget is always saturated,
 - some good projects always remain unfunded.

The agents

Entrepreneurs

A project needs a sunk cost $I \in (0, 1)$, yields an outcome in $\{0, 1\}$ with success probability π . Density f , c.d.f. F . No effort cost: the entrepreneur needs no equity.

The agency

- Budget B_a
- Knows π for each project
- Profit-maximizing

Investors

- Budget B_i
- Know only the distribution of π
- Competitive (price-taking)

Allocation $\mu = (\sigma, \rho)$: the agency contributes $\sigma(\pi)$ to the cost I and keeps a share $\rho(\pi)$ of the outcome; the investor completes (and takes) the rest.

Payoffs and assumptions

Payoffs. For each project π :

$$U_a(\pi) = \rho(\pi)\pi - \sigma(\pi) \quad U_i(\pi) = (1 - \rho(\pi))\pi - (I - \sigma(\pi))$$

Budget constraints. $B_a \geq \int \sigma(\pi)f(\pi) d\pi$ and $B_i \geq \int (I - \sigma(\pi))f(\pi) d\pi$

Assumptions

- **(A1)** $\mathbb{E}(\pi) - I < 0 < 1 - I$, and f log-concave: the average project is not worth financing, the best one is.
- **(A2)** $B_a < I(1 - F(I)) \leq B_a + B_i$: the agency cannot finance all good projects alone, but *together* they can.
- **(A3)** The agency is profit-maximizing.

Game 1 (Benchmark)

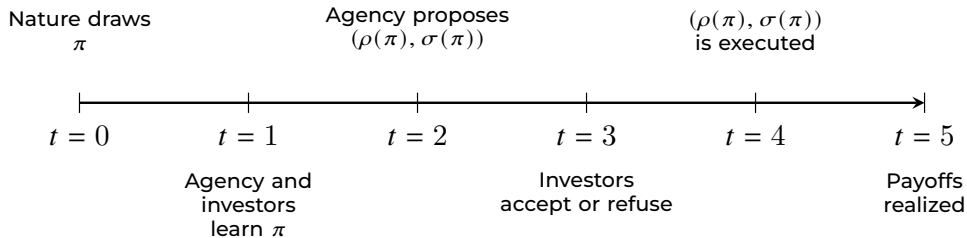


Figure: Game 1 – Benchmark

Game 1 (Benchmark)

Proposition 1

Under budget and investors participation constraints:

The agency maximizes profit	$\Leftrightarrow$	Investor utility $U_i(\pi) = 0$ for each π The agency finances all good projects
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Investors participation constraint: $U_i(\pi) \geq 0$ for each π

Game 1 (Benchmark)

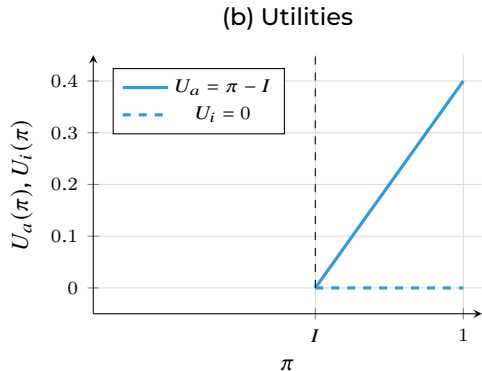
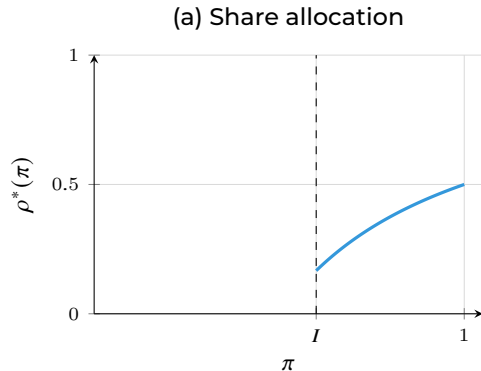


Figure: Parameters: $\pi \sim U[0, 1]$, $I = 0.6$, $\sigma^* = 0.1$. (a) The share $\rho^*(\pi) = 1 - (I - \sigma^*)/\pi$ is increasing in π . (b) The agency captures the full surplus $\pi - I$; the investor earns zero rent.

Game 2 (Ex-post commitment)

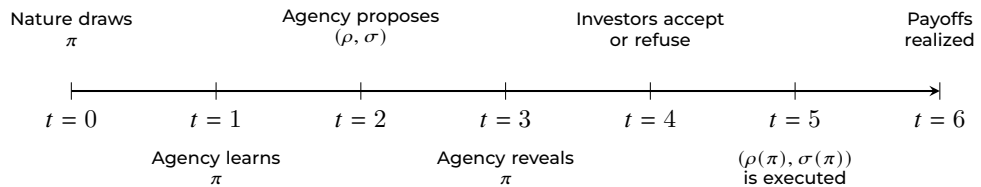


Figure: Game 2 – Ex-post commitment

Game 2 (Ex-post commitment)

Proposition 2

Under budget and investors participation constraints:

The agency
maximizes profit

$\Leftrightarrow$

The allocation is pooling

The agency saturates its budget

$U_i(\pi_{\min}) = 0$ for the less good
financed project

The profit-maximization FOC is
satisfied

Investors participation constraints:

- $U_i(\pi) \geq 0$ for each π
- $U_a(\pi) \geq \rho(\hat{\pi})\pi - \sigma(\hat{\pi})$ for each $(\pi, \hat{\pi})$

Game 2 (Ex-post commitment)

Remark

$\pi_{\min} > I$: Some good projects always remain unfunded.

A contract for each project

Four projects, $I = 1/2$, $B_a = 0.4$. Tailored terms, every investor exactly at zero.

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	$(0.1, 0.05)$	$(0.1, 0.05)$	$(0.4, 0.05)$	$(0.6, 0.1)$	
Agency U_a	0	0	0.25	0.50	$\Sigma = 0.75$
Investor U_i	0	0	0	0	$\Sigma = 0.00$

The agency takes the whole surplus, 0.75, and spends 0.25 of its budget.

The agency is the one who says which project is which

It reports a $\pi = 1/2$ project as $\pi = 1$: $0.6 \cdot \frac{1}{2} - 0.1 = 0.20 > 0$.

	reported as $\pi = 1$				
	▼				
Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	(0.1, 0.05)	(0.6, 0.1)	(0.4, 0.05)	(0.6, 0.1)	
Agency U_a	0	0.20	0.25	0.50	$\Sigma = 0.95$
Investor U_i	0	-0.20	0	0	$\Sigma = -0.20$

Every project below $\pi = 1$ wants that contract, and the constraint $U_i(\pi) \geq 0$ is written project by project.

One contract for all, and a saturated budget

The best single contract is $(0.6, 0.2)$: it funds $\pi \geq 3/4$ and spends $2 \times 0.2 = B_a$.

Project	$\pi = \frac{1}{2}$	$\pi = \frac{1}{2}$	$\pi = \frac{3}{4}$	$\pi = 1$	
Contract (ρ, σ)	not funded	not funded	$(0.6, 0.2)$	$(0.6, 0.2)$	
Agency U_a	—	—	0.25	0.40	$\Sigma = 0.65$
Investor U_i	—	—	0	0.10	$\Sigma = 0.10$

Pooling, budget saturated, $U_i(\pi_{\min}) = 0$ — and $\pi_{\min} = 3/4 > I = 1/2$.

Game 2 (Ex-post commitment)

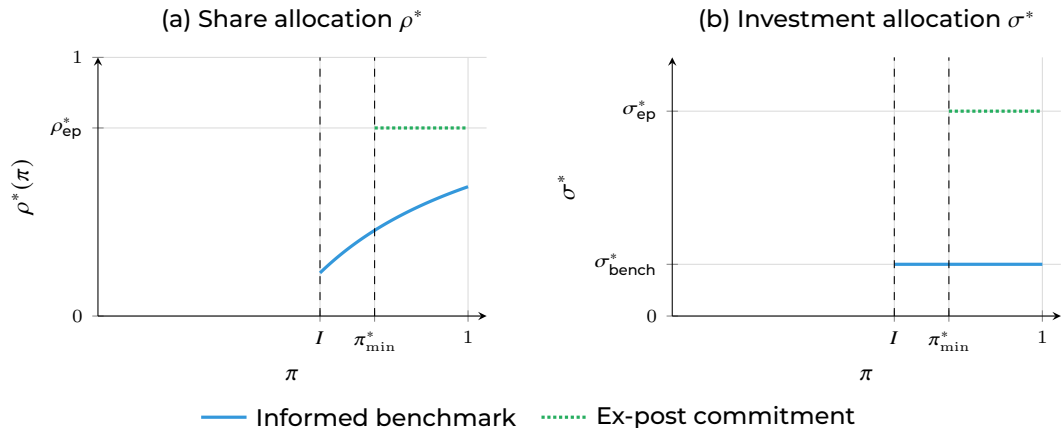


Figure: Informed benchmark vs. ex-post commitment regime. Parameters: $\pi \sim U[0, 1]$, $I = 0.6$, $B_a = 0.1$, $B_i = 0.2$.

Game 2 (Ex-post commitment)

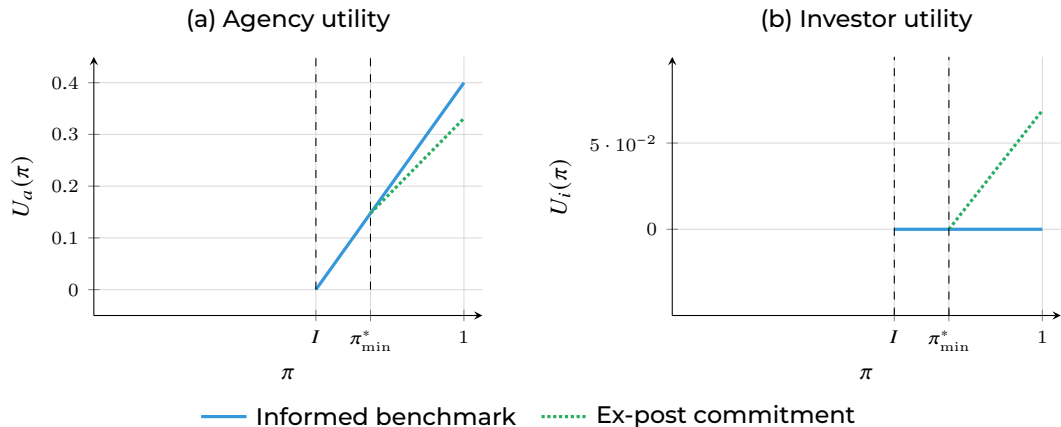


Figure: Utilities per project: informed benchmark vs. ex-post commitment regime. Same parameters.

Game 3 (Ex-ante commitment)

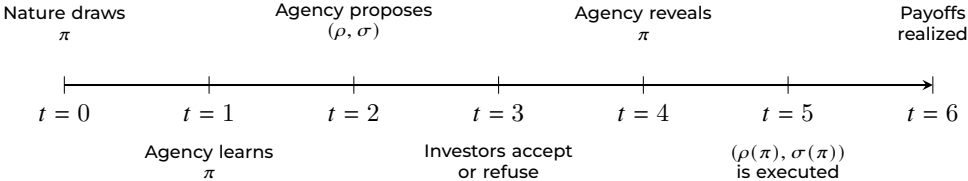


Figure: Game 3 – Ex-ante commitment

Game 3 (Ex-ante commitment)

Proposition 3

Under budget and investors participation constraints:

The agency
maximizes profit



The allocation is pooling

The agency does not saturate its
budget

Investor average utility

$$\int U_i(\pi) f(\pi) d\pi = 0$$

The agency finances all good
projects

Investors participation constraint: $\int U_i(\pi) f(\pi) d\pi \geq 0$

Game 3 (Ex-ante commitment)

Remark

The reverse is not true.

Game 3 (Ex-ante commitment)

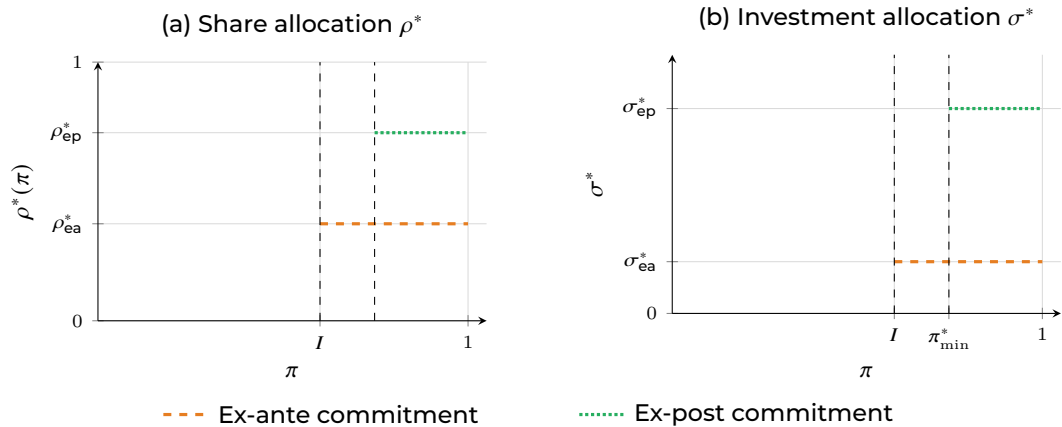


Figure: Optimal allocation under ex-ante vs. ex-post commitment investors. Same parameters as in Game 2.

Game 3 (Ex-ante commitment)

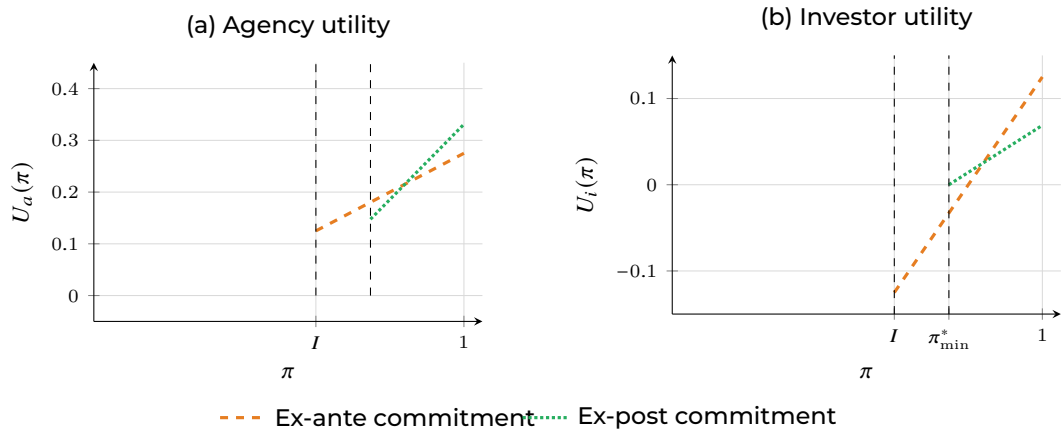
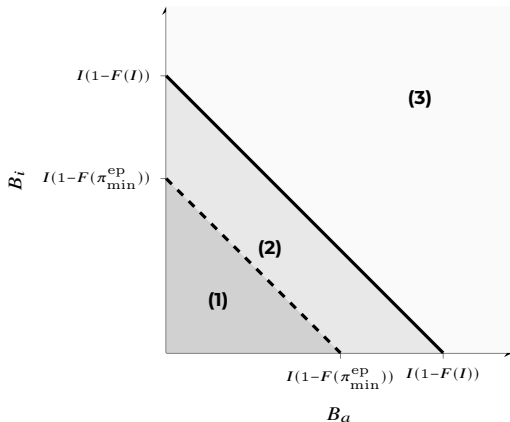


Figure: Utilities under the two commitment regimes. Same parameters as in Game 2.

Cash-poor vs. cash-rich economies

The friction is slack only where capital itself is short.



- (1) Severely cash-poor: both regimes coincide
- (2) Moderately cash-poor: ex-post excludes more
- (3) Cash-rich: ex-post excludes $[I, \pi_{\min}^{\text{ep}})$

Where public money should enter

The state as a limited partner, not as an accelerator.

Financing threshold	Public accelerator <i>ex-post commitment</i>	Public LP <i>ex-ante commitment</i>
(3) Cash-rich	$\pi_{\min}^{\text{ep}} > I$	I
(2) Moderately cash-poor	$\pi_{\min}^{\text{ep}}$	$(1 - F)^{-1} \left(\frac{B_a + B_i}{I} \right) < \pi_{\min}^{\text{ep}}$
(1) Severely cash-poor	$(1 - F)^{-1} \left(\frac{B_a + B_i}{I} \right)$	$(1 - F)^{-1} \left(\frac{B_a + B_i}{I} \right)$

EIF and Bpifrance already operate this way.

Main result

- Ex-ante: pooling is optimal, but not uniquely.
 - Ex-post: pooling is unique, budget-saturating, inefficient.
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- **Pooling rationalized.** Optimal between VC institutions.
 - **The discipline device.** A binding budget buys selection.
 - **Policy.** Public capital belongs at the LP stage, unless capital itself is short.

Thank you
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